

Option Pricing and Hedging – Monte Carlo Methods

1 Introduction

In this project I used Monte Carlo methods to price and hedge a range of options. I began by pricing European options using both the Black-Scholes formula and Monte Carlo simulation, then priced American options using the Longstaff-Schwartz algorithm, before extending the Monte Carlo approach to Asian and knock-in barrier options. To improve the efficiency of the simulations I applied antithetic variates, control variates and importance sampling. I then computed the Greeks and used them to simulate delta hedging under proportional transaction costs.

In the final section, I implemented the Whalley-Wilmott no-transaction band, which is a closed-form asymptotic approximation to the optimal hedging policy under transactional costs. I then tested numerically the cube-root scaling law between transaction cost rate and optimal band width. The objective is to demonstrate both practical implementation and theoretical understanding of numerical methods used in derivative pricing and risk management.

2 Pricing Models

2.1 Setup

I first imported one year of historical Apple stock price data to compute the annualised volatility. This was done by computing the daily log returns $\log(C_t/C_{t-1})$ for $t = 1, \dots, 249$, where C_t is the closing price at time t . I then computed the standard deviation of the daily log returns and multiplied by $\sqrt{250}$ to obtain the annualised volatility, with 250 trading days assumed as weekends and holidays reduce the year to approximately that Figure [1]. All simulations assume a constant risk-free rate of 4.09%, corresponding to the 1-Year U.S. Treasury Constant Maturity Rate for May 2025, and a constant expiry of one year. The latest closing price in the imported data was used as the initial stock price S_0 .

For simplicity, volatility is treated as constant and known throughout this project, i.e. the same σ is used to price the option, compute the hedge ratio, and to generate simulated stock paths. Dividends are also ignored.

2.2 Black-Scholes Formula

The Black-Scholes model describes the evolution of the stock price through

$$dS(t) = rS(t) dt + \sigma S(t) dW(t)$$

[2, page 4] where $S(t)$ is the stock price at time t , $W(t)$ is a standard Brownian motion, σ is the volatility and r is the risk-free rate. The solution to this equation leads to the following closed-form prices:

- $d_1 = \frac{\log(S_0/K) + [r + \frac{1}{2}\sigma^2] T}{\sigma\sqrt{T}}$
- $d_2 = d_1 - \sigma\sqrt{T}$
- Call price = $S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$
- Put price = $Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)$

where Φ is the standard normal distribution function.

2.3 Monte Carlo Simulation

I then priced the same option by Monte Carlo simulation [2, page 5]:

- $S_i(T) = S_0 \exp\left(\left[r - \frac{1}{2}\sigma^2\right]T + \sigma\sqrt{T}Z_i\right)$ for $Z_i \sim N(0, 1)$, $i = 1, \dots, n$, where n is the number of simulations,
- $C_i = e^{-rT}(S_i(T) - K)^+$,
- $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$ is the estimated European call price.

The estimator is unbiased and converges at rate $O(n^{-1/2})$ by the central limit theorem, so halving the standard error requires four times as many paths. With sufficient simulations the two methods agree, as shown in Figure 1.

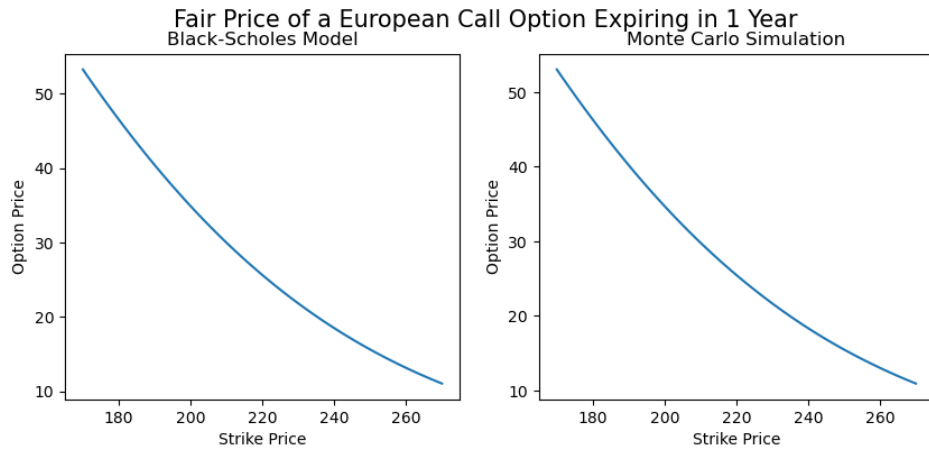


Figure 1: Black-Scholes and Monte Carlo prices across strikes

2.4 Longstaff-Schwartz Algorithm

An American option may be exercised at any time before expiry, so the holder faces an optimal stopping problem. The time to maturity is split into steps and the option is valued at each. The value of an American call at time t is $\max((S_t - K)^+, C_t)$ where C_t is the continuation value. To estimate the continuation value, least squares regression is used to approximate the expected cash flow from keeping the option alive, conditional on the current stock price [3]. Let X be the stock prices at time t for which the option is in the money, and Y the corresponding discounted cash flows received at time $t+1$ if the option is not exercised at t . Regressing Y on X and X^2 gives an estimate of $E[Y | X]$. Exercise is taken to be optimal where the immediate exercise value exceeds the fitted continuation value, and cash flows are adjusted on those paths only. The recursion runs backwards from the terminal payoff to the first step.

Restricting the regression to in-the-money paths is done because out-of-the-money paths carry no exercise decision, and including them adds noise to the fit without adding information. Results are shown in Figure 2, using 100,000 simulations and 52 steps.

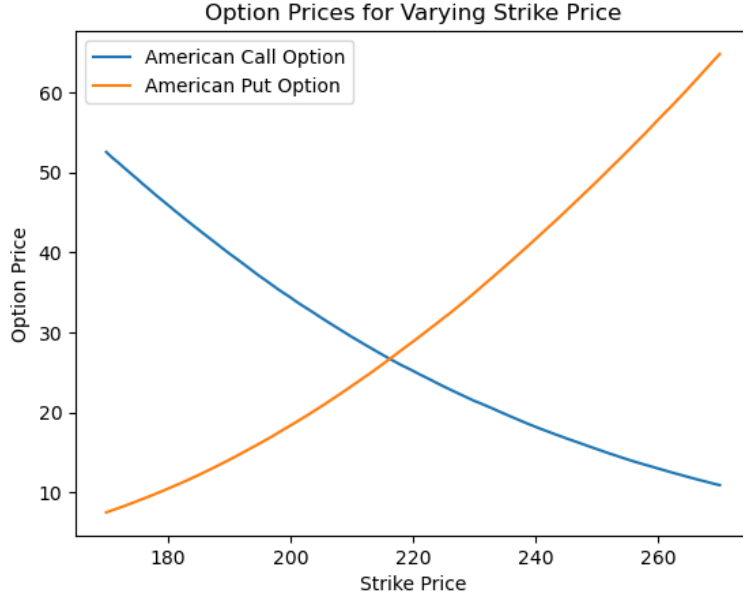


Figure 2: American call and put prices across strikes

As expected, the American call price coincides with the European price, since early exercise of a call on a non-dividend-paying stock is never optimal. The American put trades at a premium to its European counterpart, reflecting the value of the early exercise right.

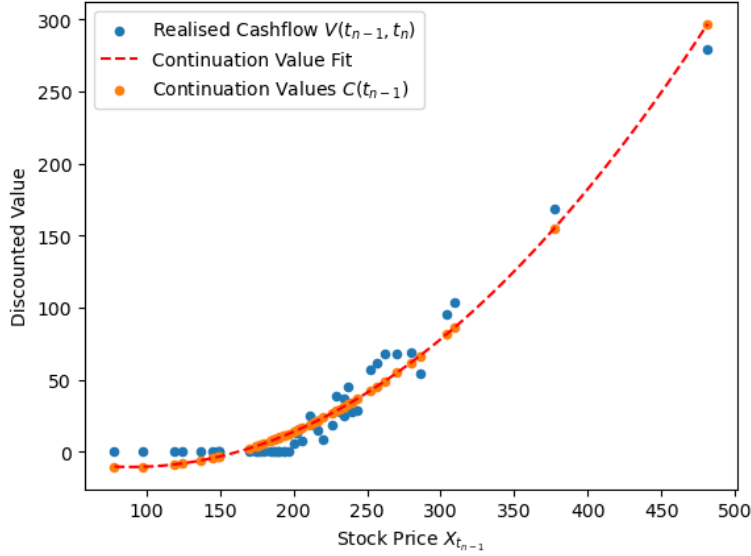


Figure 3: Polynomial Estimate for Continuation Values

Figure 3 shows the regression at a single exercise date: the blue points are realised discounted cash-flows plotted against the prior stock price, the red dashed line is the fitted quadratic, and the orange points are the resulting continuation value estimates at each in-the-money path.

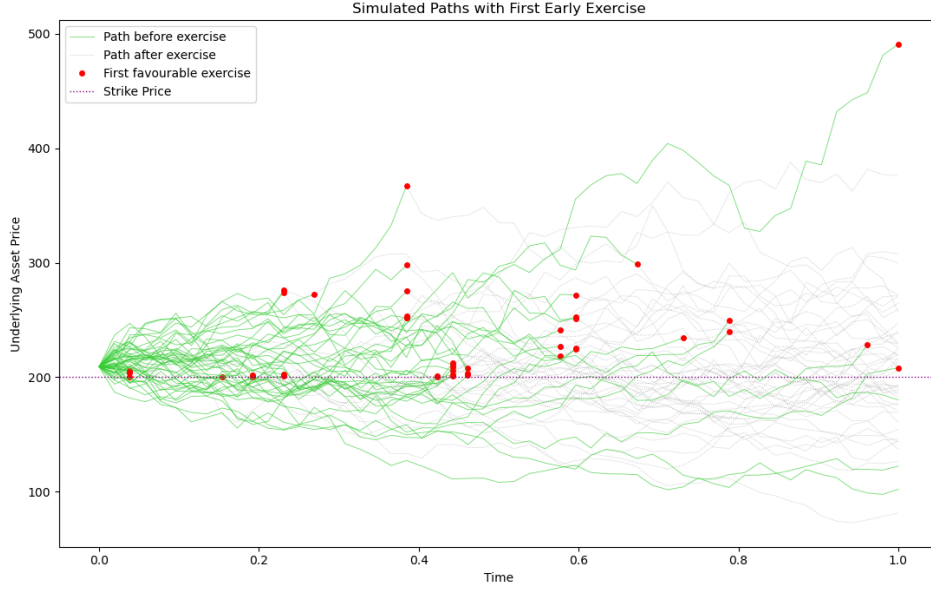


Figure 4: Simulated Paths with First Early Exercise

Figure 4 shows simulated paths for the American call, coloured green before and grey (dotted) after each path's first favourable exercise, with red markers indicating the exercise point.

2.5 Exotic Options

For an Asian call the payoff at expiry is $(S_{\text{avg}} - K)^+$, where S_{avg} is the mean stock price over the monitoring dates, excluding time 0; for a put it is $(K - S_{\text{avg}})^+$. Because the average is required, the stock price must be simulated over steps as for the American option, though pricing is cheaper since no regression is needed. Averaging reduces the influence of the final price, resulting in Asian being less volatile and thus cheaper than their European equivalents.

For a knock-in barrier option the payoff matches a European option but requires the barrier being touched at some time in $(0, T]$. It must be touched from below for a call or from above for a put [4]. The path must therefore be simulated so that the barrier condition can be checked at each step. The barrier is checked only at simulation steps rather than continuously, so a path that crosses the barrier between two steps and returns is recorded as never having crossed. The estimator is therefore biased downwards.

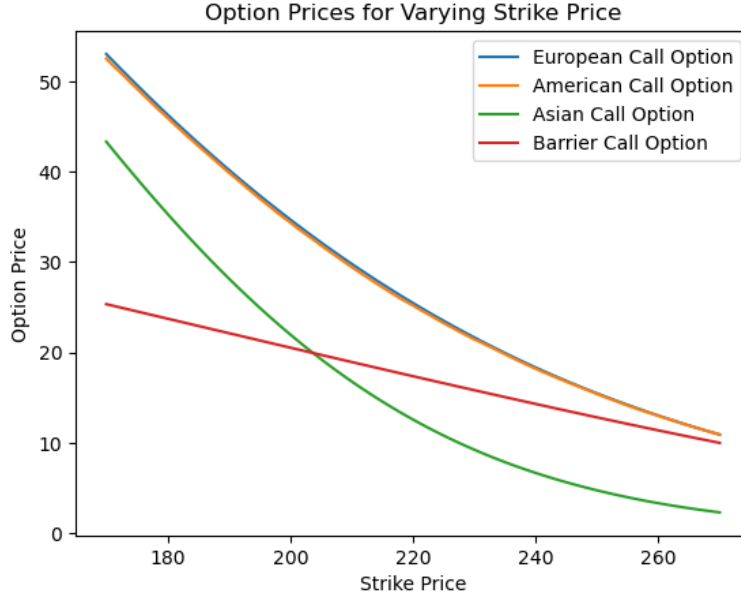


Figure 5: Call prices across strikes for each option type

3 Variance Reduction

3.1 Antithetic Variates

The method of antithetic variates pairs $Z_1, Z_2, \dots$ i.i.d. $N(0, 1)$ with $-Z_1, -Z_2, \dots$ when simulating paths [2, page 205]. The two payoffs in a pair are negatively correlated, so their average has lower variance than the average of two independent draws. Writing Y_i and $\tilde{Y}_i$ for the payoffs from Z_i and $-Z_i$, the estimator is the sample mean of the pair averages $\frac{1}{2}(Y_i + \tilde{Y}_i)$, whose variance is

$$\frac{1}{2} \left(\text{Var}(Y) + \text{Cov}(Y, \tilde{Y}) \right),$$

so the technique helps exactly when the covariance term is negative. Applied to a European call with 100,000 simulations, the results are shown in Figure 6.

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Price of Option without Variance Reduction: 25.6
Price of Option with Antithetic Variates: 25.57

Variance of Price Estimator without Variance Reduction: 0.010836
Variance of Price Estimator with Antithetic Variates: 0.007512
Variance reduction: 30.68%
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Figure 6: Results from antithetic variates

The reduction obtained is modest, with the antithetic estimator achieving a 30.68% reduction in the variance of the price estimator using 100,000 simulations, and prices of 25.60 and 25.57 respectively, consistent with an unbiased estimator.

3.2 Control Variates

Control variates use a quantity with known expectation to correct the Monte Carlo error of a related quantity. Let Y_i be the Asian payoff and X_i the geometric Asian payoff generated on the same path,

with $E[X]$ given exactly by the Black-Scholes formula. The adjusted payoff is

$$Y_i(b) = Y_i - b(X_i - E[X])$$

[2, page 186], which is unbiased for any b . The variance-minimising choice is

$$b^* = \frac{\text{Cov}[X, Y]}{\text{Var}[X]},$$

estimated from the sample, and the resulting variance reduction is exactly ρ_{XY}^2 , the squared correlation between the two payoffs. Results using 100,000 simulations are shown in Figure 7. The geometric Asian is a valid control because its price is known analytically. The geometric average of log-normal prices is itself log-normal, so the contract prices by a Black-Scholes-type formula, whereas the arithmetic average is not log-normal does not admit a closed-form solution

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Price of Option without Variance Reduction: 12.68
Price of Option with Control Variates: 12.68

Variance of Price Estimator without Variance Reduction: 0.005623
Variance of Price Estimator with Control Variates: 0.0001
Variance reduction: 98.23%

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Figure 7: Results from control variates

The two payoffs differ only in the choice of mean and are correspondingly highly correlated, giving a variance reduction of 98.23% and an implied correlation of $\rho \approx 0.991$. This is equivalent to running roughly 57 times as many simulations at no additional cost.

3.3 Importance Sampling

Importance sampling was used to reduce the variance of knock-in barrier payoffs by shifting the sampling distribution so that the rare barrier-crossing event becomes common. The increments are drawn from $N(\mu, 1)$ rather than $N(0, 1)$, and payoffs are re-weighted by the likelihood ratio

$$\frac{L(0)}{L(\mu)} = \frac{\prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}Z_i^2\right)}{\prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(Z_i - \mu)^2\right)} = \exp\left(-\mu \sum_{i=1}^n Z_i + \frac{n}{2}\mu^2\right)$$

[2, page 255], where n is the number of steps and the Z_i are the shifted draws. The re-weighting leaves the estimator unbiased while concentrating simulation effort on the paths that actually contribute to the payoff. I used a shift of $\mu = 0.3$ with 100,000 simulations.

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Price of Option without Variance Reduction: 8.66
Price of Option with Importance Sampling: 8.49

Variance of Price Estimator without Variance Reduction: 0.015487
Variance of Price Estimator with Importance Sampling: 0.001178
Variance reduction: 92.39%

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Figure 8: Results from importance sampling

This produced a variance reduction of 92.39%. This is large because the original estimator wastes most of its paths as most never reach the barrier. The estimator is dominated by a small number of paths.

4 Greeks

The Greeks are defined as

- Delta: $\Delta = \frac{\partial V}{\partial S}$
- Gamma: $\Gamma = \frac{\partial^2 V}{\partial S^2}$
- Vega: $\mathcal{V} = \frac{\partial V}{\partial \sigma}$
- Rho: $\rho = \frac{\partial V}{\partial r}$
- Theta: $\Theta = \frac{\partial V}{\partial t}$

where V is the option price, S the current stock price, σ the volatility, r the risk-free rate and t time. I computed these using finite differences, since this extends without modification to the exotic payoffs, for which no closed-form Greeks exist.

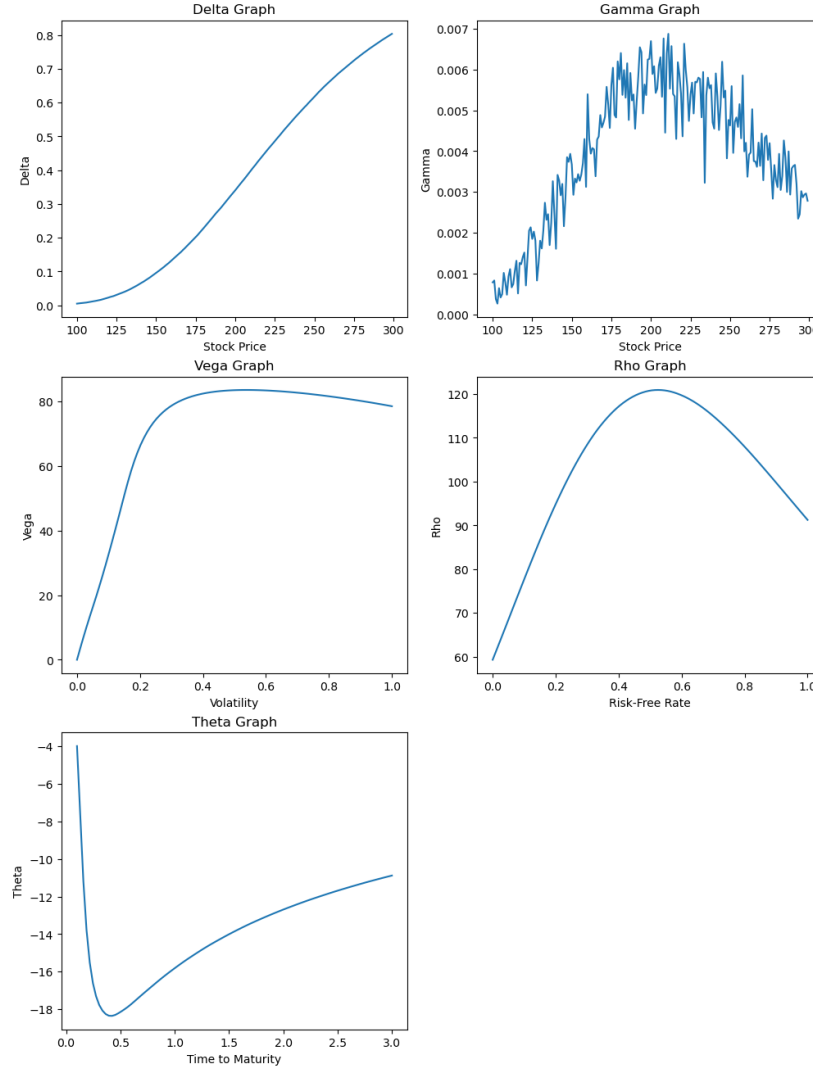


Figure 9: Greeks for a European call option

The Gamma curve is visibly noisier than the others, though the values remain below 0.012 throughout. This is a consequence of the second-order difference: the estimator divides by ϵ^2 rather than ϵ , so any residual Monte Carlo error in the three valuations is amplified by a factor of ϵ^{-2} . There is a bias-variance trade-off in the choice of bump size, with discretisation bias falling and variance rising as ϵ decreases. For the European call the closed-form expressions

$$\Delta = \Phi(d_1), \quad \Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T}}$$

are available and were used inside the hedging simulations of the following sections, both for accuracy and because the finite-difference estimator would otherwise require a nested simulation at every time step of every path.

5 Delta Hedging with Transaction Costs

I simulated delta hedging a short European call option. The hedge holds Δ shares against the written option, where Δ is the Black-Scholes delta. Each step (trading day), the stock price is updated, Δ recomputed, and the share position moved to match it. Cash accumulates interest at the risk-free rate. At expiry the shares are sold and the option settled against its payoff.

Rebalancing every step would be prohibitively expensive once proportional costs are charged on each trade, so the position is only adjusted when Δ has moved by at least a threshold ϵ . Larger ϵ means fewer trades and lower costs, but a hedge that tracks the option less closely. I recorded profit and loss across simulated paths for a range of ϵ ; results are in Figure 10.

Epsilon	Mean P&L	S.E.	Std	2.5%	97.5%
0.001	-2.29	0.04	1.38	-5.37	-0.33
0.05	-0.90	0.06	1.85	-4.83	2.91
0.1	-0.64	0.11	3.61	-7.81	6.28
0.2	-0.80	0.24	7.54	-16.85	12.54
0.5	-0.42	0.50	15.69	-31.07	26.02
1	0.11	0.70	22.26	-52.80	27.69

Figure 10: Profit and loss for varying ϵ

As ϵ increases the mean profit and loss moves towards zero while the dispersion widens substantially. This is the expected behaviour: fewer trades mean lower cumulative transaction costs and hence a smaller mean loss, but a hedge that tracks delta less closely and therefore a wider distribution of outcomes.

A methodological point worth recording is that all values of ϵ were evaluated on the same set of simulated paths. Using common random numbers in this way removes path-to-path variation from the comparison, so that the differences between rows reflect the effect of the threshold alone.

6 Delta Hedging with the Whalley-Wilmott Band

A constant threshold applies the same tolerance regardless of how fast the delta is moving. Near expiry and close to the strike, delta shifts sharply for small moves in the stock, so a fixed band triggers far more often than it does when the option is deep out of the money. A better policy would adapt to the state of the option, and one follows from treating the problem as utility maximisation.

6.1 The no-transaction band

Hodges and Neuberger [5] treat hedging with transaction costs as a utility maximisation problem, using the exponential utility $U(W) = -e^{-\gamma W}$ where γ measures risk aversion. Their solution gives a

band around the Black-Scholes delta. No trading happens while the holding stays inside it, and the hedge is only adjusted when it leaves. The band has to be found numerically.

Whalley and Wilmott [6] approximated this band for small transaction costs, giving the closed form

$$h = \left(\frac{3}{2} e^{-r\tau} \frac{k S \Gamma^2}{\gamma} \right)^{1/3},$$

where k is the cost rate, S the stock price, Γ the option gamma and τ the time remaining. The hedge is rebalanced when $|\Delta_{\text{held}} - \Delta_{\text{BS}}| > h$. A high gamma means delta moves quickly, so rebalancing would happen often and cost more, and the band widens to compensate.

I used this in place of the constant threshold, computing Γ from its closed form at each step. I swept γ over a logarithmic grid, since the band only depends on it through $\gamma^{-1/3}$ and so changes slowly. Results are in Figure 11.

Gamma Risk	Mean P&L	S.E.	Std	2.5%	97.5%
1e-05	0.05	0.46	14.43	-29.62	26.76
0.0001	-0.76	0.35	11.09	-26.30	24.45
0.001	-0.79	0.20	6.42	-14.79	11.07
0.01	-0.57	0.10	3.19	-7.26	5.67
0.1	-0.84	0.06	1.88	-5.18	2.48
1	-1.37	0.04	1.32	-4.48	0.59

Figure 11: Profit and loss across risk-aversion levels

The sweep shows the same trade-off as before. Low risk aversion gives a wide band, so few trades are made and the results are more spread out, while high risk aversion gives the opposite. At the high end the results match those from the smallest constant thresholds, which is what should happen since both approach continuous hedging, and confirms the implementation is consistent.

6.2 Testing the cube-root scaling

The $1/3$ exponent comes from balancing two effects. Delta crosses a band of width h about h^{-2} times, each trade costing around kSh , so total costs scale as k/h . A wider band also lets the hedge drift further from the correct delta, so the variance scales as h^2 . Maximising

$$J(h) = E[\Pi] - \lambda \text{Var}[\Pi]$$

balances k/h against λh^2 , giving $h \propto k^{1/3}$.

To test this, I swept ϵ over a logarithmic grid for each cost rate k and found the ϵ maximising J , fitting a quadratic around the peak so the answer was not pinned to grid points. Fitting a line to $\log \epsilon$ against $\log k$ then gives the exponent directly.

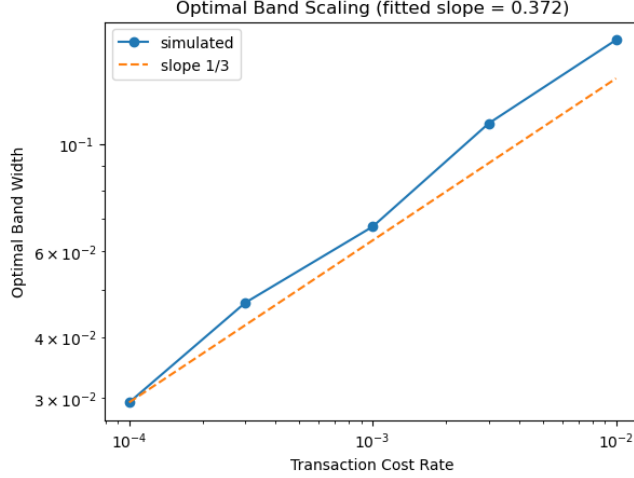


Figure 12: Optimal band width against transaction cost rate, logarithmic axes

The fitted exponent was 0.372, close to the theoretical $1/3$ but not equal to it. The gap comes from the time discretisation. Between steps the delta moves by roughly $\Gamma S \sigma \sqrt{\Delta t}$, around 0.04 at the money with 100 steps per year, so any band narrower than this is crossed almost every step and behaves the same as continuous hedging. At low cost rates the true optimum falls below that limit, so the measured optimum is too large and the fitted slope too steep. Repeating the sweep with more steps per year moved the exponent closer to $1/3$, confirming the discretisation as the cause. Very large cost rates fail for the opposite reason, as the optimal band covers nearly the whole range of delta and the objective flattens, so I restricted the sweep to the intermediate range.

7 Limitations and Further Work

Several simplifications constrain the interpretation of these results.

Volatility is constant, known, and identical across pricing, hedging and simulation. As noted in Section 2.1, this forces the mean hedging profit and loss to be negative and equal to the transaction cost drag, and it removes any exposure to volatility risk. The natural extension is to price and hedge at an implied volatility distinct from the volatility used to generate paths, which recovers the standard decomposition

$$\Pi \approx \frac{1}{2} \int_0^T \Gamma_t S_t^2 (\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2) dt,$$

under which delta hedging is revealed as a bet on implied against realised volatility rather than a risk-free operation. Implementing this requires only that the two volatilities be separated in the simulation; interpreting it against market data would require observed option prices, which were not available.

Stock paths are generated under geometric Brownian motion, so returns are log-normal with constant volatility. Real equity returns exhibit heavier tails, volatility clustering and jumps, all of which worsen hedging performance and none of which the present simulation can capture. Re-running the hedge over overlapping windows of historical returns, or under a stochastic volatility model such as Heston, would quantify the resulting model risk.

Only delta is hedged. Under a stochastic volatility model the residual exposure to Vega would become a material source of hedging error, and neutralising it would require trading other options rather than the underlying alone.

Finally, the Whalley-Wilmott band was compared against constant thresholds only through the cube-root scaling of the latter. A direct comparison of the two policies on a common mean-variance

frontier, at a fixed cost rate, would test the state dependence of the band itself, which is the distinctive content of the result and which the scaling test does not address.

References

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